MMCS Final Project
Optimization Models for a Future Edinburgh Cycle Scheme

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1 Introduction

While carbon neutrality and sustainability have been widely discussed globally, as well as in Scotland, the shift towards cycling has become an attractive option for achieving such goals. To avoid falling behind in global decarbonisation efforts, it is crucial to revolutionise the transport system as a solution (Hussaini & Scholz, 2017). In response, in this study, we suggest a bike-sharing system as one of the instruments contributing to such goals, focusing specifically on the operational optimisation perspective. Edinburgh's narrow streets face traffic congestion and safety concerns that can degrade residents' quality of life (Hart & Parkhurst, 2011), making bike-sharing system a potential solution to mitigate these problems.

Edinburgh has experience of the failure of a bike-sharing programme with Just Eat Cycle from 2018 to 2021. Serco managed this project under a self-financing model without public funding. They used dock-based station systems that have relatively low risks of vandalism or theft. However, their failure came from substantial initial investment and ongoing expenditure for station installation, permits, power supply, communication system setup, and long-term maintenance. Particularly in cities with limited urban space like Edinburgh, these costs carry a high risk of deficit. Meanwhile, Voi Technology AB launched its electric bicycle system in Edinburgh in September 2025 entirely on private capital, separate from the City of Edinburgh Council's budget. The key difference is its dockless system and utilizing geofencing-based virtual parking spaces. The dockless system offers relatively low initial installation costs and flexibility to scale up or down according to demand fluctuations. Furthermore, the business plans to expand its deployment to about 800 bicycles by the end of 2025. A well-constructed optimisation model with low computational costs is a key element of success to verify demand, supply, and cost variables in real time.

Shared bicycle schemes cannot be defined as either purely market-based services or purely public services. In other words, a service combining public objectives with private profitability is required, one with both economic and sociological goals. Therefore, the primary operational objective is not simply to deploy large numbers of bicycles but to select the optimal number of bicycles and potential demand areas, taking into account budget, urban demand characteristics, and equity. While a larger number of bicycles increases the choices available to citizens, it does not guarantee the sustainability of the business (in terms of maintenance costs and budget design). Furthermore, deploying large bicycle stations may improve citizens' accessibility, but it can create long-term risks for the business in terms of cost recovery due to imbalances between demand and supply. In essence, maximising demand from the perspective of citizens and the city itself conflicts with minimising costs that represent the operator's interests. In other words, shared bicycle operations must be carefully considered from economic, urban planning, and mathematical perspectives to enhance both citizen convenience and operator sustainability.

Therefore, we conduct a multidisciplinary strategy to optimise the number of bicycles and the locations of virtual stations within Edinburgh's Dockless Electric Bicycle System by applying Mixed-Integer Linear Programming (MILP), considering limited budget constraints. Particularly, given that a high level of dense stations in the past hindered mobility for vulnerable groups such as the elderly or disabled, identifying a central point by reducing unnecessary stations is essential. To reflect these characteristics and neutralise outliers in demand maps, candidate sites are selected based on historical demand using k-means clustering. Subsequently, regional demand is empirically forecasted to estimate each station's potential demand. An MILP model is formulated, incorporating budget, bicycle count, railway station coverage, equity, distance constraints, forecasted demand, and network connectivity. This scenario considers administrative, economic, and public characteristics.

Our result contributes to providing quantitative decision-making resources for the company and city government. Specifically, the firm, utilizing its own capital, must collaborate with the city for long-term viability. Establishing a compelling pivoting strategy and securing corporate support funds and subsidies should be future objectives. These insights can be demonstrated through budget setting, determining the optimal number of business bicycles, establishing the bicycle fleet size, and formulating budget adjustment scenarios. The results demonstrate that shared bicycles have become a significant strategic resource for cities and businesses, transcending their perception as merely an additional mode of transport.

2 Background and Literature Review

Edinburgh is a historical and cultural centre and a major strategic location in Scotland. This section provides background information and insight into prior research for bike-sharing businesses in Edinburgh.

The substitution of bicycles as a mode of transport offers benefits in reducing carbon emissions, alleviating traffic congestion, and promoting eco-friendly travel (Macagno, 2013; Sandor, 2017). Existing studies emphasise the integration of bike-sharing services with urban infrastructure and public transport (Hickman & Banister, 2007; Pinto et al., 2019), which can revolutionise citizens' mobility patterns and contribute to public health by encouraging physical activity (Hoj et al., 2018; Fishman & Cherry, 2016; Jiang et al., 2020).

However, rapid growth created problems including oversupply, indiscriminate parking, and encroachment on pedestrian space (Tu et al., 2019; Hua et al., 2020; Wang et al., 2020). Large-scale abandonment resulted in the "bicycle graveyard" phenomenon in several cities. Recent dockless schemes address these issues through geofencing-based virtual parking zones, photo verification, penalties, and coordination with local authorities (Sun et al., 2019).

2.1 Bicycle Deployment Problem

Selecting bike station location considers potential demand factors such as population, activity levels, and public transport stops [García-Palomares et al., 2012]. The maximal coverage location problem (MCLP) maximises demand covered within a given number of facilities (Church & ReVelle, 1974) and is particularly suitable for bike-sharing stations (Frade & Ribeiro, 2015). Yuan et al. (2019) apply MILP to constrain station location, capacity, and redeployment. Frade and Ribeiro (2015) present an MCLP model that maximises demand within budget constraints. Demand estimation utilizes variables such as population density, worker density, and facility type (Garcia-Palomares et al., 2012). Interconnection with public transport is significant, as station locations benefit from integration with public transport networks (Garcia-Palomares et al., 2012; Yuan et al., 2019).

The issue of selecting bike station locations has been considered based on potential demand factors such as population, activity levels, and public transport stops (García-Palomares et al., 2012). These problems involve multiple factors, each subject to practical constraints. Consequently, attempts have been made to identify station locations using optimization tools such as location-allocation models (García-Palomares et al., 2012). Particular attention should be paid to effectively utilizing the complementary influence zones of transfer stations and bus stops.

2.2 Dockless System

Following the mid-2010s, the advancement of IT technology alongside GPS and smartphones led to the emergence of dockless shared bicycle systems (Chen et al., 2020). While previous station-based systems faced rigidity in licensing and site selection, dockless technology significantly reduced infrastructure and licensing costs. This offers the advantage of rapid city-wide expansion and greater flexibility than previous systems. The introduction of smartphone apps, real-time location tracking, and surveillance-based systems has enabled the pre-emptive mitigation of risks such as theft (Graham et al., 2014). Particularly in China, companies like Mobike and ofo deployed vast numbers of bicycles within a short timeframe after 2015 (Say et al., 2018).

3 Modelling Approach

A Mixed Integer Linear Programming Model was used for this problem with the following parameters and decision variables.

Model Formulation		
Index sets		
I		The set of all locations
T		The set of train stations
Decision Variables		
y_i	$\{0,1\}$	Whether to build a station at location i or not
x_i	$\mathbb{Z}_0^+$	The number of bikes allocated to location i
$a_{i,j}$	$\{0,1\}$	Indicating if location i is allocated to location j
covered_t	$\{0,1\}$	Indicating if train station t is near some bike station
Parameters		
dist_{ij}	$\mathbb{R}_0^+$	The distance from location i to location j
$\text{dist}^{\min}$	$\mathbb{R}_0^+$	The minimum allowable distance between any two stations, 0.4km
$\text{dist}^{\max}$	$\mathbb{R}_0^+$	The maximum allowable distance between two "nearby" stations, 1km
$\text{bikes}^{\max}$	$\mathbb{Z}_0^+$	The maximum number of bikes at a station, 30
$\text{bikes}^{\text{Total}}$	$\mathbb{Z}_0^+$	The total number of bikes allowed in the solution, 800
d_i	$\mathbb{R}_0^+$	The estimated demand in location i
b_t^{station}	$\mathbb{R}_0^+$	The extra demand satisfied for building near train station t
c_i^{bike}	$\mathbb{R}_0^+$	The purchase cost and 1 year of maintenance of one bike, £1,500
c_i^{station}	$\mathbb{R}_0^+$	The purchase cost and 1 year of maintenance of one station, £5,000
B	$\mathbb{R}_0^+$	The total project budget, £1,500,000

Table 1: Notation for the Mixed Integer Linear Programming Model

3.1 Locations

Candidate locations for bikes were chosen from Points of Interest (POIs) across Edinburgh and the surrounding area this is the set I in Table 1. A significant computational factor is the number of candidate locations, $|I|$. Solving a model with $|I|$ beyond 600 proved prohibitively time-consuming on personal laptops and so a value of $|I| = 400$ was chosen.

Initially about 33,000 Points of Interest were considered; for the above computational reasons, a subset of 400 locations were chosen by K-means clustering and formed I. Specifically, we employed weighted K-means to form 400 clusters, with higher weights given to commercial areas, hospitals and universities, which were under-represented in the POI dataset. For each of the 400 clusters, we retained the POI in each cluster which was closest to the cluster centroid - this ensured that the locations are in fact a subset of the original physical locations.

3.2 Demand

We define demand in a location as the potential additional trips by bike that could be made from that location if there were enough bikes available. We estimated demand for a location using a combination of past historical data, nearby landmarks and bicycle traffic data. The estimated average daily demand for a given location i is given by the weighted sum:

$$d_i = \alpha_1 z_i^1 + \alpha_2 z_i^2 + \alpha_3 z_i^3 \qquad \alpha_1 + \alpha_2 + \alpha_3 = 1 \tag{1}$$

Where in (1):

1. z_i^1 is the estimate using Poisson regression based on the number and types of other POIs in the vicinity. A location with many other facilities nearby may become a popular commuter hub.

2. z_i^2 is the estimate using historical trip data. We attributed the historical demand from the previous bicycle pilot to the nearby locations in this scheme. It is sensible to incorporate data the previous pilot that ran from 2018 to 2021, as it gives a recent sample of user demand where there were bicycle stations previously.

3. z_i^3 is the estimate using a small neural network trained on traffic data in Edinburgh. Public traffic statistics provide the average daily bicycle traffic at junctions in Edinburgh. From this we can use Machine Learning models to infer the bicycle traffic at each location in our model. With this α_3 captures the unrealised demand as a percentage of existing bicycle traffic.

The details and choice of weights are given in Appendix A.3.

3.3 Mixed Integer Programming Model

This can be expressed in a mixed integer programming model with the objective of maximising demand. Below are the parameters, objective function, decision variables, and constraints used with respective explanations.

3.3.1 Basic Station and Bike Constraints

We avoid the issue of rebalancing in this model due to its complexity and view it as a fixed cost that can be absorbed into the values of c_i^{bike} and c_i^{station}. Due to practical considerations, there is a limit on the total number of bikes used in the solution, $\text{bikes}^{\text{Total}}$ and that can be stored at a station, $\text{bikes}^{\text{max}}$. Bikes can only be placed in a location if a station is built there and we will only build in a location if we will place at least one bike there.

$$\forall i \in I \quad x_i \leq \text{bikes}^{\text{max}} y_i \tag{2}$$

$$\forall i \in I \quad y_i \leq x_i \tag{3}$$

$$\forall i \in I \quad \sum_{i \in I} x_i \leq \text{bikes}^{\text{Total}} \tag{4}$$

We ensure that we do more bikes for which there is demand for at a station. The plus one term in the following constraints allows for the case where there is no demand at a location but it may allow for connectivity between stations– connectivity is discussed in Section 3.3.2.

$$\forall i \in I \quad x_i \leq d_i + 1 \tag{5}$$

We need to stay under budget given the fixed costs of bikes and stations.

$$\sum_{i \in I} c_i^{\text{bike}} x_i + c_i^{\text{station}} y_i \leq B \tag{6}$$

To prevent overcrowding in popular areas, we will not build two stations in locations that are within dist^{min} of each other. We calculated the distance between locations i and j, d_{ij}, as GPS distance using the longitude and latitude of the locations- this is a simplification, as the physical distances as travelled by bicycle will typically be larger.

$$\forall i \in I \quad \text{dist}^{\text{min}} y_i \geq \sum_{j \in I} d_{ij} y_j \tag{7}$$

3.3.2 Connectivity Constraints

We ensure that the stations built form a connected network, or possibly multiple small connected networks. The motivation is to ensure each station is near to another station so that trips by the user are not untenably long. We take inspiration from the approach in Kit et al. (2025), which models network connectivity as an allocation problem. Specifically, we have the binary decision variable a_{ij} which indicates if a location i is allocated to location j. We then view each location $i \in I$ as a node in a graph and each allocation, a_{ij}, as an edge connecting two nodes. We restrict our allocations to only the locations where we build stations and enforce that we cannot simultaneously allocate two locations to each other through the following constraint.

$$\forall i, j \in I \quad a_{ij} + a_{ji} \leq y_i \tag{8}$$

We require that each node has another node allocated to it, i.e. each station has another station near it, and enforce that the in-degree of each node is one. This leads to spoke patterns in the network which is acceptable as some stations may connect users from the outskirts to more central areas.

$$\forall i \in I \quad y_j = \sum_{j \in I} a_{j,i} \tag{9}$$

We want to prevent cycles in the graph of length less than 4 as such subnetworks were considered too small to properly provide users with realistic travel options. For a given subset $S \subset I$ where $|S| < 4$, we enforce the constraint:

$$\sum_{i,j \in S} a_{ij} \leq \sum_{i \in S} y_i - \left(\sum_{i \in S} y_i \right) / |S| \tag{10}$$

There is $C_4^{|I|}$ subsets of I which are of size four and so applying the constraint (10) would lead to an explosion in the number of constraints. Rather we first solve the model without any constraints of the form (10), then iteratively identify cycles (of length less than four) in the network and apply the constraint in (10) to the nodes in those cycles and then resolve and iterate until we find a solution with no such cycles.

We will only allocate a station to another station within $\text{dist}^{\max}$ distance. This is equivalent to specifying a maximum edge length of $\text{dist}^{\max}$ - note that we do not specify a minimum edge length, as this is captured by (7) and (8) jointly.

$$\forall i \in I \quad \text{dist}^{\max} y_i \geq \sum_{j \in I} d_{ij} a_{ji} \tag{11}$$

3.3.3 Train Station Coverage Constraints

We model the train station coverage as a reward within the objective function, where a train station t is covered if we build at least one station y_i within 400 meters:

$$\forall t \in T \quad \text{covered}_t \leq \sum_{i \in I \text{ and } d_{it} \leq .4} y_i \tag{12}$$

We allow for different rewards per train station, which can be interpreted as additional daily trips by users in a last-mile connection, b_t^{station}.

3.3.4 Objective Function

First, we allocate bicycles x_i based on the estimated demand d_i to maximise the total utility. Furthermore, our solution aims to complement the current transport network; as such, we aim to cover each of the major railway stations within Edinburgh, similar to that of Garcia & Palomares et al. (2012). In which we use an additional reward term b_t^{station} if train station t is covered, covered_t, by a bike station. Therefore, our objective function is as follows:

$$\text{maximise} \quad \sum_{i \in I} d_i x_i + \sum_{t \in T} b_t^{\text{station}} \text{covered}_t \tag{13}$$

4 Result

4.1 Estimating an Initial Budget

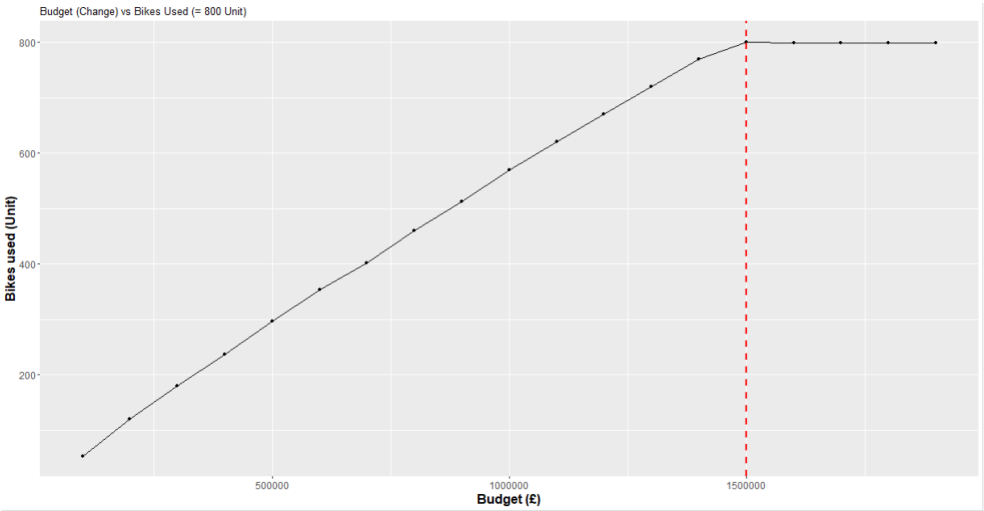


Figure 1: Budget Constraint Sensitivity Analysis

Given Voi currently has a bike-sharing pilot scheme, we use their plan for the total number of bikes to be deployed in order to estimate an appropriate initial budget. A substantial budget allocation expands the permissible scope of managerial decisions at the project's inception, but an excessively large budget risks significant financial waste. Therefore, it is vital to select a reasonable budget allocation point aligned with the business expansion objectives. We calculate the reasonable budget required to deploy up to 800 bikes ($\text{bikes}^{\text{Total}}$) in Edinburgh, considering the assumed basic operational expenditure (purchase costs, maintenance costs), and then identify the blueprint locations.

Figure 1 shows the budget required to deploy up to 800 bicycles in Edinburgh, adjusting the budget in increments of £100,000. In practice, from a budget of £100,000 up to £1.4 million, the number of bicycles allocated increases as the budget increases. This is an obvious fact, as a larger budget allows for the purchase of more bicycles. However, the allocation peaks at £1.5 million, with a similar number of bicycles being allocated thereafter. That is, £1.5 million is appropriate for deploying the initial target of 800 bicycles, and budgets exceeding this range may be inefficient for sustaining the operation. Figure 2 shows the optimal station placement we identified for an input budget of £1.5 million.

4.2 Main Result

Our solution is presented below, with the left image showing the full optimal solution and the right providing a magnified view of a connectivity example. Red circles indicate proposed bike station locations, with circle size representing station capacity. Blue markers show the train stations around which we strategically positioned our network, while connecting lines illustrate bike station connectivity. We have allocated 800 bikes across 55 dockless stations within the £1.5 million budget. The implementation is designed as a phased strategy, beginning with a smaller pilot launch and scaling up to meet maximum demand. This approach allows for observation and real-data collection, enabling us to rebalance station locations and bike quantities as needed—a key strength of our flexible model. By positioning stations near train stations, we aim to create mobility synergy with public transport, providing a last-mile solution for commuters. Our solution establishes four main community networks: a central network which extends from the Southside to Leith, one covering Portobello, one along Peffermill Road and the final one covering the business park at South Gyle. Despite these networks being, in theory, disconnected, they are still within cycling distance of one another and are also connected by the main cycle roads across Edinburgh.

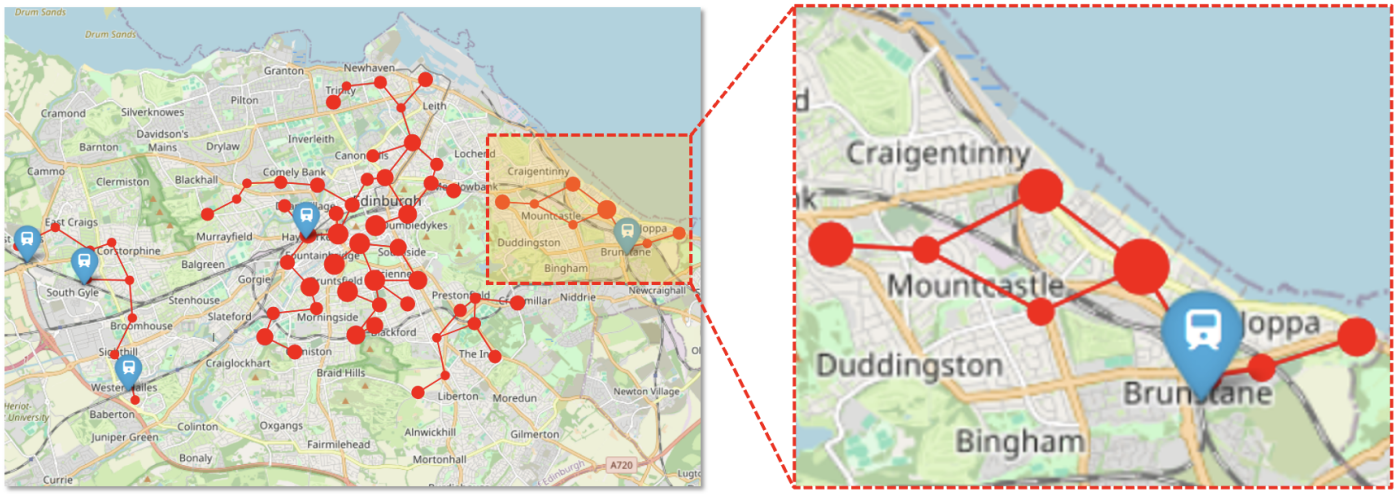


Figure 2: Main Result (Optimal Location with Connectivity)

4.3 Implementation by Phase

Allocating all 800 bicycles from the outset, as per the business objective, exposes the company to significant operational risk when demand locations remain uncertain. By establishing a smaller initial entry point and gradually expanding, the company can mitigate losses from unforeseen risks.

Therefore, we propose a phased approach for implementation. This study conducts scenario analysis by incrementing the budget in £0.5 million units, from £0.5 million to £1.5 million.

In the first phase, a budget of £500,000 is used to construct nine stations, meeting 12% of total demand. The second phase further expands the station network, focusing on the city centre and establishing a community within Portobello. This phase, with a budget of £1 million, meets 26% of demand. The final phase, with a total budget of £1.5 million, meets 35% of demand.

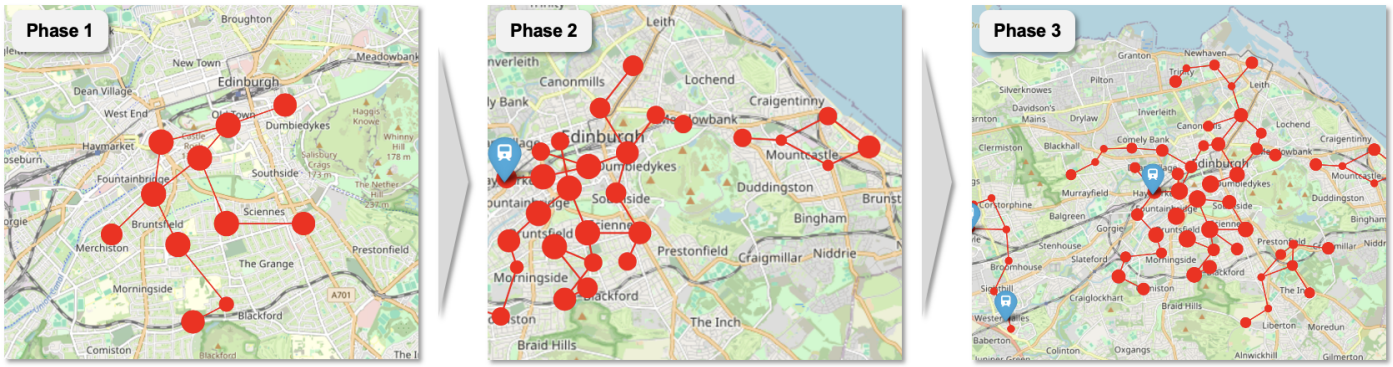


Figure 3: Implementation by 3 Phases

When a company's target investment budget is determined, it can proceed with investments across various demand areas by dividing them into multiple phases. Our results demonstrate an optimal example of how a company can allocate a £1.5 million budget across three phases for bicycle stations. This approach simultaneously identifies each station's connectivity, proximity to public transport (train stations), and high-demand areas, enabling potential customers to be guided through three distinct phases.

Furthermore, our findings represent a plan based on current conditions, considering only limited factors. After deploying bicycles at the initial pilot location and securing actual customer demand data, the firm can estimate this through a pilot study. This is an important feature of our model due to the inherent uncertainty within the demand estimation; as such, we can respond and reoptimise in response to receiving more certain demand and customer feedback during the phased rollout.

4.4 Proportion of Daily Demand Covered

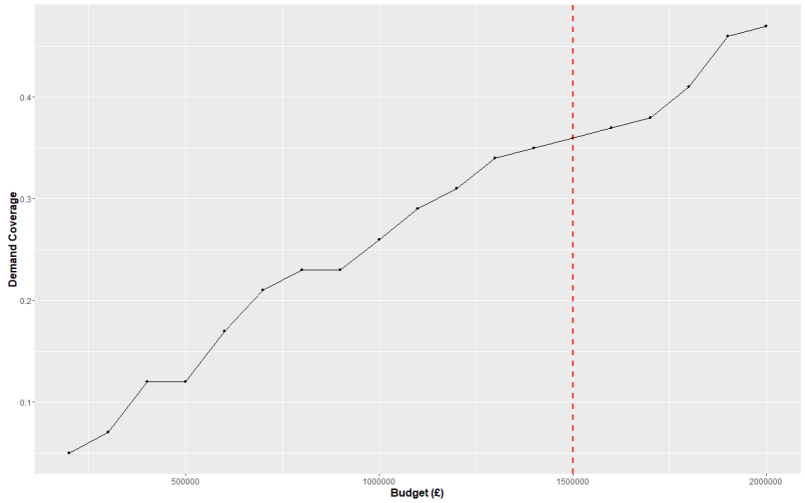


Figure 4: Sensitivity of Demand Coverage to the Total Budget

Given the direct relationship between the budget and the number of stations, we examine the sensitivity of the budget to the total demand covered. We resolved the model, holding all other parameters constant and increasing the budget from £100,000 to £2,000,000. The result shows that the demand covered is largely a function of the size of the budget. This allows the council to select an appropriate budget with respect to their required demand coverage.

It is important to note that coverage does not increase linearly with the budget. This non-linearity is driven by the connectivity constraints; high-demand areas cannot be served in isolation unless there are sufficient intermediate stations to connect them to the network or a sufficient number of stations to create their own community. As a result, our solution may appear to underperform alternative solutions in terms of demand coverage. However, in our view there is no value in covering demand on paper if the resulting stations are disconnected. Such solutions would result in reduced ridership and frustration for users.

5 Conclusion

This study examines the reintroduced bicycle sharing scheme in Edinburgh and proposes a MILP model to determine the optimal number of bicycles and station locations within a constrained budget. Our results comprehensively consider multiple factors such as uncertain user demand, public transport (train station) linkages, and the need for plausible bicycle networks.

Edinburgh previously experienced the failure of Just Eat Cycles. Dock-based stations imposed significant initial costs and may have made the scheme vulnerable to fluctuations in demand due to the rigidity of location changes. The model and scenario analysis in this study consider dockless-based stations. Dockless stations can readily relocate to respond immediately to demand fluctuations and have lower initial start-up costs than dock-based stations. They also offer sustained labour cost savings in maintenance. Therefore, drawing lessons from past failures, this study proposes a suitable strategic deployment structure for transitioning to dockless systems. When deploying 800 bicycles, our results indicate that £1.5 million represents the optimal initial investment budget. Furthermore, a phased investment strategy, allocating the budget in £0.5 million increments, provides a decision-making approach capable of adapting to changing demand. Moving forward, our model allows for continuous recalibration of station locations capacities based on actual customer data collection. Such recalibration is significantly more appealing under the dockless approach as there is much less investment lost if a station proves to be unviable, as compared to a dock-station. In this regard, we consider our approach robust under uncertainty in user demand. An additional advantage of the solution is that it provides realistic travel connections through a network of nearby bicycle stations and linkages with existing public transport infrastructure.

References

[1] Searle, K. D., Searle, C., & Kalcsics, J. (2025). *Infrastructure design for a hydrogen refuelling network with pipelines. Journal of the Operational Research Society, 1-22.*

[2] Yuan, M., Zhang, Q., Wang, B., Liang, Y., & Zhang, H. (2019). *A mixed integer linear programming model for optimal planning of bicycle sharing systems: A case study in Beijing. Sustainable cities and society, 47, 101515.*

[3] Say, A. L., Guo, R., & Chen, C. (2018, August) *Disruption or new order?: The emergence of the unicorn bike-sharing entrepreneurship in China. In 2018 Portland International Conference on Management of Engineering and Technology (PICMET) (pp. 1-9). IEEE.*

[4] Chen, Z., Van Lierop, D., & Ettema, D. (2020). *Dockless bike-sharing systems: what are the implications? Transport reviews, 40(3), 333-353.*

[5] Hussaini, M., & Scholz, M. (2017). *Exploring low carbon transition pathways for the UK road transport sector. Transportation Planning and Technology, 40(7), 796-811.*

[6] Wang, Y., Jia, S., Zhou, H., Charlton, S., & Hazen, B. (2021). *Factors affecting orderly parking of dockless shared bicycles: an exploratory study. International Journal of Logistics Research and Applications, 24(2), 103-125.*

[7] Sun, L., Zhou, X., & Sun, Z. (2019). *Improving cycling behaviors of dockless bike-sharing users based on an extended theory of planned behavior and credit-based supervision policies in China. Frontiers in psychology, 10, 2189.*

[8] Pinto, R., Espírito-Santo, A., & Paciello, V. (2019, October). *Proposal of a sustainable e-Bike sharing infrastructure based on the IEEE 1451 standard. In IECON 2019-45th Annual Conference of the IEEE Industrial Electronics Society (Vol. 1, pp. 5538-5543). IEEE.*

[9] Tu, Y., Chen, P., Gao, X., Yang, J., & Chen, X. (2019). *How to make dockless bikeshare good for cities: Curbing oversupplied bikes. Transportation Research Record, 2673(6), 618-627.*

[10] Hart, J., & Parkhurst, G. (2011). *Driven to excess: Impacts of motor vehicles on the quality of life of residents of three streets in Bristol UK. World Transport Policy & Practice, 17(2), 12-30.*

[11] Frade, I., & Ribeiro, A. (2015). *Bike-sharing stations: A maximal covering location approach. Transportation Research Part A: Policy and Practice, 82, 216-227.*

[12] Church, R., & Velle, C. R. (1974). *The maximal covering location problem. Papers in regional science, 32(1), 101-118.*

[13] García-Palomares, J. C., Gutiérrez, J., & Latorre, M. (2012). *Optimizing the location of stations in bike-sharing programs: A GIS approach. Applied Geography, 35(1-2), 235-246.*

[14] Woodcock, J., Tainio, M., Cheshire, J., O'Brien, O., & Goodman, A. (2014). *Health effects of the London bicycle sharing system: health impact modelling study. Bmj, 348.*

[15] Macagno, T. (2013). *A model for managing corporate sustainability. Business and Society Review, 118(2), 223-252.*

[16] Sandor, R. L. (2017). *The Benefits of Corporate Sustainability. World Scientific Book Chapters, 220-223.*

[17] Hickman, R., McCartney, P., & Banister, D. (2007). *Transport and global warming: what is the potential for carbon reduction in Scotland?..*

[18] Hoj, T. H., Bramwell, J. J., Lister, C., Grant, E., Crookston, B. T., Hall, C., & West, J. H. (2018). *Increasing active transportation through e-bike use: pilot study comparing the health benefits, attitudes, and beliefs surrounding e-bikes and conventional bikes. JMIR public health and surveillance, 4(4), e10461.*

[19] Fishman, E., & Cherry, C. (2016). *E-bikes in the Mainstream: Reviewing a Decade of Research. Transport reviews, 36(1), 72-91.*

[20] Jiang, Z., Lei, C., & Ouyang, Y. (2020). *Optimal investment and management of shared bikes in a competitive market. Transportation research part B: methodological, 135, 143-155.*

[21] Tu, Y., Chen, P., Gao, X., Yang, J., & Chen, X. (2019). *How to make dockless bikeshare good for cities: Curbing oversupplied bikes. Transportation Research Record, 2673(6), 618-627.*

[22] Wang, Y., Jia, S., Zhou, H., Charlton, S., & Hazen, B. (2021). *Factors affecting orderly parking of dockless shared bicycles: an exploratory study. International Journal of Logistics Research and Applications, 24(2), 103-125.*

[23] UK Road traffic statistics, Road Traffic in Great Britain (2024) *https://roadtraffic.dft.gov.uk/*

[24] Open Data Scotland, Just Eat Cycles-Edinburgh (2018-2021) *https://usmart.io/org/cyclingscotland/discovery/disco view-detail/8550147c-9ce2-4382-929d-3b4ab43e554f*

A Appendix

A.1 Repository

The source code can be found: https://github.com/michaeltiller/MMCS-Main-Project.

A.2 Point of Interests

Table 2 summaries the Points of Interest (POIs) dataset used initially to generate candidate locations. The POIs are categorised broadly by the type of area they are in (commercial/residential) or the amenity they are in the vicinity of (Hospital/School/Library/University). Within the weighted K-means, we assign a weight of 3 to university and commercial areas, a weight of 2 to schools and hospitals, and a weight of 1 to libraries and residential areas.

Points of Interest	
Category	**Count**
Residential	32,837
Commerical	551
Hospital	16
School	188
Library	49
University	28
Total	**33,669**

Table 2: Categorisation of the full Points of Interest Dataset.

A.3 Demand Estimation

We use (1) with the weights tuned to $\alpha_1 = 0.4, \alpha_2 = 0.2, \alpha_3 = 0.4$ to estimate user demand at the candidate locations. The choice of weights was found by tuning so that the predictions gave realistic results upon inspection of selected Points of Interest e.g. Haymarket Station and Portobello Beach.

We now further elaborate on the three components of our estimate; nearby Points of Interest, historical demand and traffic data prediction.

A.3.1 Regression using nearby Points of Interest

The count of the yearly historical trip starts from the previous Edinburgh bike scheme and is used to train a Poisson Generalised Linear Model (GLM). For each former station $k \in K$, we count POIs from categories $C = \{\text{commercial}, \text{hospital}, \text{library}, \text{residential}, \text{school}, \text{university}\}$ within buffer distances $D = \{100, 250, 500, 700 \text{ m}\}$. Let $X_{k,dc}$ be the number of POIs of category c within distance d of station k, and define $\mathbf{x}_k = (X_{k,dc})_{d \in D,\, c \in C}$. The model is:

$$y_k^{(s)} \sim \text{Poisson}(\lambda_k^{(s)}), \qquad \log \lambda_k^{(s)} = \beta_0^{(s)} + \mathbf{x}_k^\top \boldsymbol{\beta}^{(s)}.$$

The yearly starting trip count for a candidate location i is predicted as:

$$z_i^{(1)} = \exp\left(\hat{\beta}_0^{(s)} + \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}^{(s)}\right).$$

A.3.2 Historical data from previous pilot scheme

Given that some of our stations are within the same region as the old, we are able to directly map this demand to our current candidate locations. This is done through nearest neighbour matching through the use of a KDTree. The use of historical demand typically acts as a regularisation, given that the methods outlined previously tend to overestimate the demand within the periphery of Edinburgh, while the centre gets underestimated; as such, by combining with the old demand, this ensures a more reasonable demand distribution from the centre to more rural areas across Edinburgh.

A.3.3 Bicycle traffic data

There is publicly available data from https://roadtraffic.dft.gov.uk/ which provides the average daily number of bicycles recorded, each year, at various traffic junctions in the UK as well as the longitude and latitude coordinates of each junction. We filtered the data to only traffic junctions in the City of Edinburgh or its surrounding region. We considered data from the years 2018 to 2024 as this was the most recent data available, excluding the years of 2020 and 2021 due to the general drop in all forms of travel in those years due to the COVID-19 pandemic. We trained a small neural network, a 2-layer feed-forward network, on the longitude and latitude coordinates of the junctions in order to predict the recorded traffic counts. Having fitted this model, we then inputted the longitude and latitude coordinates of each of our candidate locations to predict the bicycle traffic at these locations.